








Statistics of Financial Markets

Seminar 9: ARCH and GARCH models



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EODHD**: EOD Historical Data (market-data provider)
- **EWMA**: Exponentially Weighted Moving Average
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **IGARCH**: Integrated GARCH
- **LR**: Likelihood Ratio (test)
- **MLE**: Maximum Likelihood Estimation
- **QLIKE**: Quasi-LIKElihood (loss function)
- **SE**: Standard Error
- **SFM**: Statistics of Financial Markets

Today's Question and Seminar Structure

- Question of the seminar
  - **how can we model, estimate and forecast a volatility that changes every day?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	persistence, long-run variance, half-life and one GARCH(1,1) step	Solved, Proposed	A1
A3, A4	multi-step forecasts, 10-day volatility and VaR 1%	Solved, Proposed	A3
A5, A6	the ARCH(1) likelihood and kurtosis; news impact in GJR-GARCH and EGARCH	Solved, Proposed	A5
B1, B2	GARCH(1,1)-t estimates: DAX; BET, Banca Transilvania, Bitcoin	Solved, Proposed	B1
B3, B4	asymmetry and the leverage effect: S&P 500; BET, Bitcoin, TLV	Solved, Proposed	B3
B5, B6	GARCH against EWMA out of sample, VaR 1% exceedances: S&P 500; DAX, BET	Solved, Proposed	B5
C1, C2	is Bitcoin volatility becoming more like equity volatility? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it after the model; the solution is discussed in the seminar
- A: calculations on paper; B: real data, in the notebook, each ending with an interpretation question; C: open questions, among them the audit of an AI answer

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, Chapter 2)
- In the notebook: `returns('dax')`, `fit(r, dist='t')` (package `arch`); no account or key is needed

Prerequisites for Today (1/17): Conditional Mean and Conditional Variance

- **Return model**: a constant mean plus a shock whose size changes over time

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t$$

  - r_t: the daily log return in %; μ: its mean; ε_t: the **shock** (the unexpected part of the return)
  - z_t: the **innovations**, i.i.d. (independent, identically distributed) with mean 0 and variance 1

- **Conditional variance**: the variance of tomorrow's return given everything known today

$$\sigma_t^2 = \mathrm{Var}(r_t \mid \text{information up to day } t-1)$$

  - σ_t^2 is known one day ahead: it is the variance forecast for day t
  - the **unconditional variance** $\mathrm{Var}(r_t)$ is the long-run average of σ_t^2
  - the shocks are uncorrelated, but their squares are correlated: volatility clustering (Chapter 8)

Prerequisites for Today (2/17): The ARCH(1) Model

- **ARCH(1)** Engle (1982) (autoregressive conditional heteroskedasticity)

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2, \qquad \omega > 0, \quad 0 \leq \alpha < 1$$

  - in words: today's variance is a constant ω plus a share α of yesterday's squared shock
  - α: the reaction to yesterday's news; $\alpha = 0$ gives a constant variance ω

- **Unconditional variance**: take expectations, with $E(\varepsilon_t^2) = E(\varepsilon_{t-1}^2) = \sigma^2$

$$\sigma^2 = \omega + \alpha\sigma^2 \quad \Rightarrow \quad \sigma^2 = \frac{\omega}{1-\alpha}$$

- **Kurtosis** $K = E(\varepsilon_t^4)/\sigma^4$ with Normal z_t and $3\alpha^2 < 1$

$$K = \frac{3\,(1-\alpha^2)}{1-3\alpha^2} > 3$$

  - $K = 3$ for the Normal distribution: ARCH produces heavy tails even with Normal innovations

Prerequisites for Today (3/17): The GARCH(1,1) Model

- **GARCH(1,1)** Bollerslev (1986): ARCH plus yesterday's variance

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0, \quad \alpha, \beta \geq 0$$

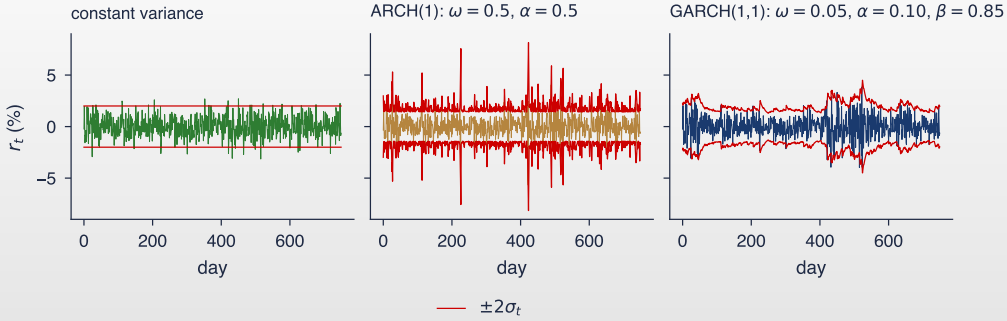
  - ω: a constant that sets the long-run level of the variance
  - α: the reaction to news (yesterday's squared shock); a large α: spiky volatility
  - β: the memory (the weight of yesterday's variance); a large β: smooth, slowly changing volatility
  - $\alpha + \beta < 1$: the model is (covariance) stationary, with a finite unconditional variance

- **EWMA** (Chapter 8) as a special case

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2$$

  - a GARCH(1,1) with $\mu = 0$, $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$, so $\alpha + \beta = 1$: an **IGARCH** (integrated GARCH)

Prerequisites for Today (4/17): Constant Variance, ARCH and GARCH in Simulation



- The same innovations z_t in the three panels; the red lines are $\pm 2\sigma_t$
- ARCH(1): the variance jumps after each large shock and falls back the next day; GARCH(1,1): the variance moves in long, smooth waves
- For a correct model, about 95% of the returns lie inside $\pm 2\sigma_t$ (Normal innovations)

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Prerequisites for Today (5/17): Persistence, Long-Run Variance and Half-Life

- **Persistence** $\alpha + \beta$ and **long-run variance**

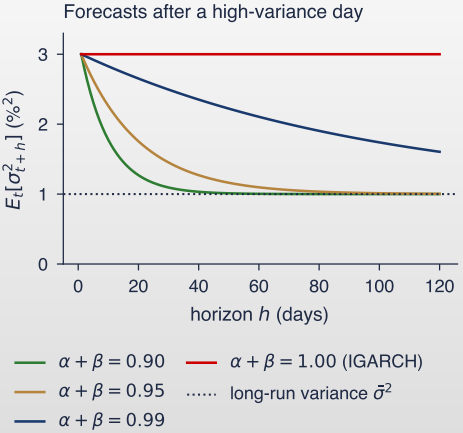
$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \qquad \sigma_{\text{year}} = \sqrt{252\,\bar{\sigma}^2}$$

  - $\bar{\sigma}^2$: the level to which the variance returns (mean reversion)
  - when $\alpha + \beta$ is close to 1, $\bar{\sigma}^2$ is very imprecise: a small error in $1 - \alpha - \beta$ changes it a lot

- Each day, the gap to $\bar{\sigma}^2$ shrinks by the factor $\alpha + \beta$

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$$

  - $h_{1/2}$: the **half-life**, the number of days after which half of a variance shock has disappeared
  - IGARCH ($\alpha + \beta = 1$): no long-run variance and no half-life; shocks never fade



Prerequisites for Today (6/17): Variance Forecasts

- Forecasts made at the end of day t; E_t: the expectation given the data up to day t
- **One day ahead**: known exactly from today's data

$$\sigma^2_{t+1} = \omega + \alpha\,(r_t - \mu)^2 + \beta\,\sigma^2_t$$

- h **days ahead**: the gap to $\bar{\sigma}^2$ decays geometrically

$$E_t[\sigma^2_{t+h}] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}\,(\sigma^2_{t+1} - \bar{\sigma}^2)$$

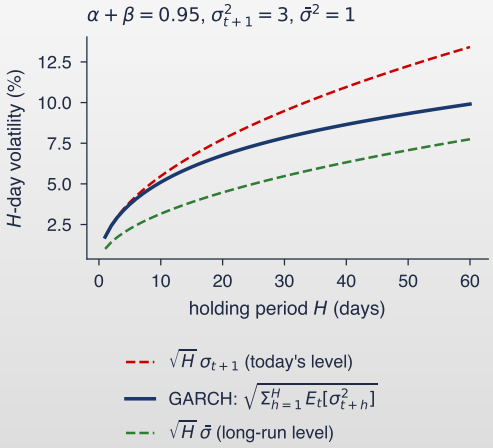
  - h: the horizon in days; for $h = 1$ the formula gives σ^2_{t+1}; for large h it tends to $\bar{\sigma}^2$

- **Variance over a holding period of H days**: the sum of the daily forecasts (returns uncorrelated)

$$\sum_{h=1}^{H} E_t[\sigma^2_{t+h}] = H\,\bar{\sigma}^2 + (\sigma^2_{t+1} - \bar{\sigma}^2)\,\frac{1-(\alpha+\beta)^H}{1-\alpha-\beta}$$

  - the second term is a geometric sum: $\sum_{h=1}^{H} q^{h-1} = (1-q^H)/(1-q)$, with $q = \alpha+\beta$
  - the H-day volatility is the square root of this sum

Prerequisites for Today (7/17): Multi-Day Risk and the Square-Root-of-Time Rule



- **Square-root-of-time rule**: $\sigma_{t:t+H} = \sqrt{H}\,\sigma$
  - exact only if the daily variance stays the same over the H days
- Two simple versions, both wrong under GARCH
  - $\sqrt{H}\,\sigma_{t+1}$: assumes today's level lasts for H days
  - $\sqrt{H}\,\bar{\sigma}$: assumes the variance is already at its long-run level
- GARCH: the square root of the sum of the daily forecasts
  - it starts at today's level and bends toward the long-run line
  - the chart uses illustrative parameters ($\omega = 0.05$, $\alpha = 0.10$, $\beta = 0.85$)

Prerequisites for Today (8/17): Value at Risk and the Innovation Quantile

- **VaR 1%** (value at risk, Chapter 6): the loss exceeded on only 1% of days

$$\mathrm{VaR}_{t+\mathbf{1}} = -\left(\mu + \sigma_{t+\mathbf{1}}\, q_{\mathbf{0.01}}(z)\right)$$

  - $q_{\mathbf{0.01}}(z)$: the 1% quantile of the innovation z, i.e. $P(z \le q_{\mathbf{0.01}}) = 0.01$; it is negative, so the VaR is a positive loss in %
  - H-day VaR: replace $\sigma_{t+\mathbf{1}}$ by the H-day volatility (previous slides) and μ by $H\mu$

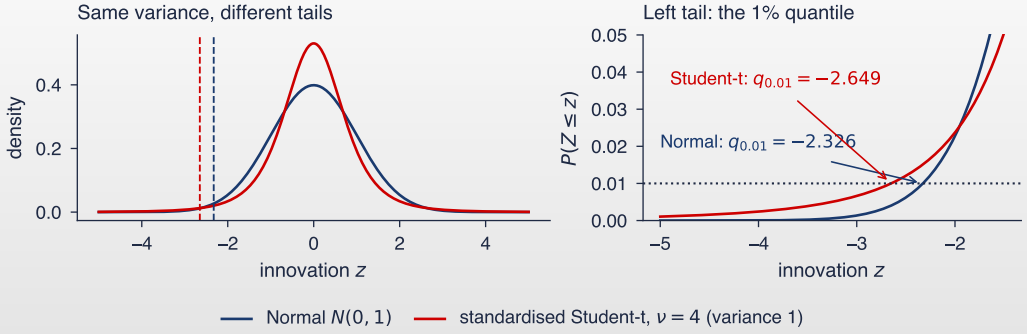
- **Normal** innovations: $q_{\mathbf{0.01}} = -2.326$

- **Standardised Student-t** innovations with $\nu > 2$ degrees of freedom Bollerslev (1987)
  - a Student-t variable t_ν has variance $\nu/(\nu-2)$; dividing by its standard deviation gives variance 1:

$$z = t_\nu \sqrt{\frac{\nu-2}{\nu}} \quad \Rightarrow \quad q_{\mathbf{0.01}}(z) = t_\nu^{-\mathbf{1}}(0.01)\sqrt{\frac{\nu-2}{\nu}}$$

  - $t_\nu^{-\mathbf{1}}(0.01)$: the 1% quantile of the Student-t with ν degrees of freedom (from tables or software)
  - a small ν: heavy tails; $\nu \to \infty$: the Normal distribution

Prerequisites for Today (9/17): Normal and Student-t Innovations



- Both densities have variance 1; the Student-t has more mass in the centre and in the tails
- Its 1% quantile is further to the left: with the same σ_{t+1}, the VaR 1% is larger

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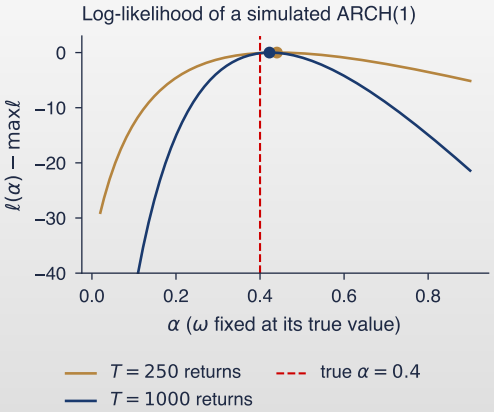
Prerequisites for Today (10/17): Maximum Likelihood

- **Conditional Gaussian log-likelihood**

$$\ell(\theta) = \sum_{t=2}^{T} \ell_t, \qquad \ell_t = -\frac{1}{2}\left[\ln(2\pi) + \ln \sigma_t^2 + \frac{(r_t - \mu)^2}{\sigma_t^2}\right]$$

  - ℓ_t: the log of the Normal density of r_t, with mean μ and variance σ_t^2; T: the number of returns
  - $\theta = (\mu, \omega, \alpha, \beta)$: the parameters; σ_t^2 is computed recursively from θ
  - a day with a large $(r_t - \mu)^2$ when σ_t^2 is small lowers ℓ_t a lot

- **MLE** (maximum likelihood estimator) $\hat{\theta}$: the θ that maximises $\ell(\theta)$
  - found numerically, under $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$
  - the chart: the more data, the sharper the peak, so the more precise $\hat{\alpha}$



Prerequisites for Today (11/17): Standard Errors and Heavy-Tailed Innovations

- **Standard errors** (SE) of $\hat{\theta}$
  - classic: from the curvature of ℓ at its maximum (a sharp peak gives small SE)
  - robust Bollerslev & Wooldridge (1992): valid also when z_t is not Normal; the `arch` package reports these
- t **statistic** of a parameter

$$t = \frac{\hat{\theta}_j}{\mathrm{SE}(\hat{\theta}_j)}$$

  - H_0: $\theta_j = 0$; approximately standard Normal under H_0; $|t| > 1.96$: significant at 5%
- **GARCH-t**: the same variance equation with standardised Student-t innovations Bollerslev (1987)
  - ℓ_t uses the Student-t density instead of the Normal one; ν is estimated with the other parameters
  - skewed t Hansen (1994): one more parameter, for asymmetric innovations

Prerequisites for Today (12/17): Comparing Models: LR Test, AIC and BIC

- **LR** (likelihood ratio) test: a larger model 1 against a nested smaller model 0

$$\mathrm{LR} = 2\,(\ell_{\mathbf{1}} - \ell_{\mathbf{0}}) \sim \chi^{\mathbf{2}}(r) \text{ under } H_{\mathbf{0}}$$

  - $\ell_{\mathbf{1}}$, $\ell_{\mathbf{0}}$: the maximised log-likelihoods; r: the number of restrictions (for example $\gamma = 0$: $r = 1$)
  - $H_{\mathbf{0}}$: the restrictions hold (the smaller model is enough); reject at 5% if $\mathrm{LR} > \chi^{\mathbf{2}}_{\mathbf{0.95}}(r)$; $\chi^{\mathbf{2}}_{\mathbf{0.95}}(1) = 3.84$

- **Information criteria** Akaike (1974), Schwarz (1978): fit penalised by the number of parameters k

$$\mathrm{AIC} = -2\,\ell(\hat{\theta}) + 2k, \qquad \mathrm{BIC} = -2\,\ell(\hat{\theta}) + k \ln T$$

  - the model with the **lower** value is preferred; BIC penalises extra parameters more ($\ln T > 2$ for $T > 7$)
  - $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathbf{1}} - \mathrm{BIC}_{\mathbf{0}} > 0$: the extra parameter does not pay for itself

Prerequisites for Today (13/17): Asymmetry: GJR-GARCH and EGARCH

- **Leverage effect** Christie (1982): falls raise volatility more than rises of the same size
- **GJR-GARCH** Glosten, Jagannathan & Runkle (1993)

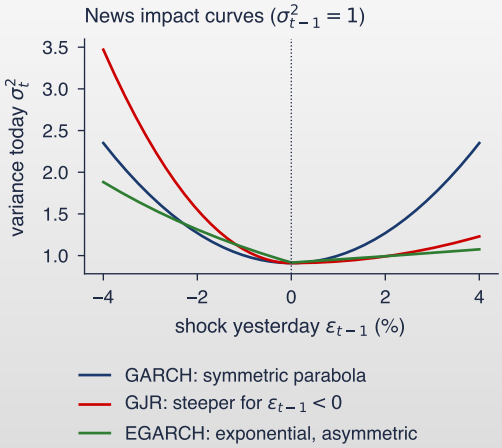
$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad I_{t-1} = \begin{cases} 1 & \varepsilon_{t-1} < 0 \\ 0 & \varepsilon_{t-1} \geq 0 \end{cases}$$

  - γ: the extra reaction to bad news; $\gamma > 0$: leverage effect
  - persistence $\alpha + \beta + \gamma/2$ for symmetric z_t (a negative shock happens half of the time)
- **EGARCH** Nelson (1991): a model for $\ln \sigma_t^2$

$$\ln \sigma_t^2 = \omega + \alpha\left(|z_{t-1}| - E|z_{t-1}|\right) + \gamma\, z_{t-1} + \beta\, \ln \sigma_{t-1}^2$$

  - $z_{t-1} = \varepsilon_{t-1}/\sigma_{t-1}$: yesterday's standardised shock; α: the size effect; γ: the sign effect, leverage if $\gamma < 0$
  - $E|z| = \sqrt{2/\pi} = 0.7979$ for Normal z; the variance $\sigma_t^2 = e^{\ln \sigma_t^2}$ is always positive

Prerequisites for Today (14/17): News Impact Curves



- **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1}, with σ^2_{t-1} fixed
  - GARCH: $\omega + \alpha\varepsilon^2_{t-1} + \beta\sigma^2_{t-1}$, a parabola symmetric around 0
  - GJR: slope coefficient $\alpha + \gamma$ on the left, α on the right
  - EGARCH: exponential in ε_{t-1}; with $\gamma < 0$, higher on the left
- Reading the chart
  - the vertical distance between the two sides at $\pm x$ measures the asymmetry
  - illustrative parameters, not estimates

Prerequisites for Today (15/17): Diagnostics of a Fitted Model

- **Standardised residuals**: the estimated innovations

$$\hat{z}_t = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}$$

  - if the model is right, $\hat{z}_t$ is close to i.i.d. with mean 0 and variance 1

- **Ljung–Box** $Q(10)$ Ljung & Box (1978) on $\hat{z}_t^2$ (Chapter 8): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k) \sim \chi^2(m)$
  - $\hat{\rho}_k$: the autocorrelation of $\hat{z}_t^2$ at lag k; no rejection: no ARCH effect left

- **Sign-bias test** Engle & Ng (1993): is there asymmetry left in $\hat{z}_t^2$?

$$\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + e_t$$

  - $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; e_t: the error of the regression
  - joint test of $b_1 = b_2 = b_3 = 0$: $TR^2 \sim \chi^2(3)$; the **sign** t: the t statistic of b_1; a rejection: a missing asymmetry

Prerequisites for Today (16/17): Out-of-Sample Comparison of Forecasts

- **Out of sample**: the forecast h_t for day t uses only data up to day $t-1$
  - the parameters are re-estimated at regular intervals (for example every 250 days) on all the data available then
  - proxy of the true variance: r_t^2 (Chapter 8)
- **QLIKE** loss Patton (2011) of day t, and its mean over N days

$$L_t = \frac{r_t^2}{h_t} + \ln h_t, \qquad \bar{L} = \frac{1}{N}\sum_t L_t$$

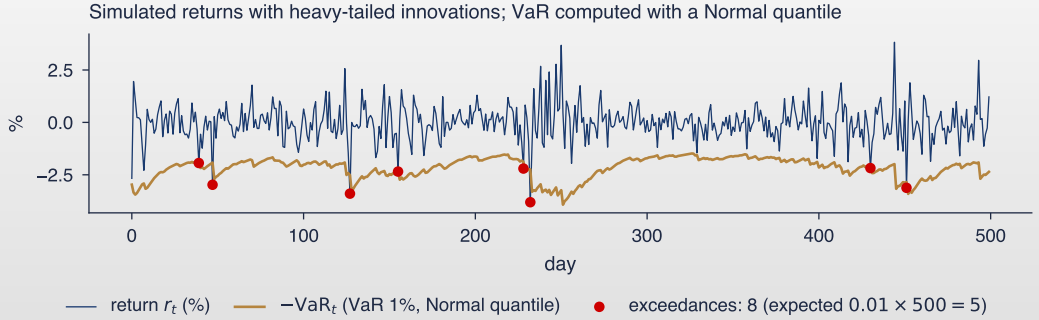
  - the smaller $\bar{L}$, the better the forecasts
- **DM** (Diebold–Mariano) test Diebold & Mariano (1995) of forecasts A and B

$$d_t = L_t^A - L_t^B, \qquad t_{DM} = \frac{\bar{d}}{\widehat{\mathrm{SE}}_{\mathrm{HAC}}(\bar{d})}$$

  - $\bar{d}$: the mean loss difference; HAC (heteroskedasticity and autocorrelation consistent): a standard error valid when d_t is autocorrelated
  - $t_{DM} < -1.96$: A is significantly better at 5%
  - **cumulative loss difference** $\sum_{s\le t}(L_s^B - L_s^A)$: rises on the days when A is better

Prerequisites for Today (17/17): VaR Exceedances

- **Exceedance** (hit): a day with $r_t < -\mathrm{VaR}_t$; indicator $I_t = 1\{r_t < -\mathrm{VaR}_t\}$ (1 if true, 0 otherwise)
- Over N days a correct VaR 1% has about $0.01\,N$ exceedances; the exceedance rate $\frac{1}{N}\sum_t I_t$ should be close to 1%; formal tests in Chapter 10



- Simulated heavy-tailed returns with a Normal VaR: the losses beyond VaR come more often than 1%
- Correlation of two volatility series (used in C1): the correlation of $\ln \hat{\sigma}_t$ of two assets on common days

A1: GARCH(1,1) Algebra [Solved]

Task

- Daily returns in %: $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, $\mu = 0$, 252 trading days per year.
- 1. Compute the persistence and the long-run variance.
- 2. Compute the annualised long-run volatility.
- 3. Compute the half-life of a variance shock.
- 4. Today $\sigma_t^2 = 1.5$ and $r_t = -3\%$; compute σ_{t+1}^2 and σ_{t+1}.
- Report: four numbers and one sentence.

Solution

- 1. $\alpha + \beta = 0.98$; $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.000$
- 2. $\sqrt{252 \times 1.000} = 15.87\%$ per year
- 3. $\ln 0.5 / \ln 0.98 = -0.69315/(-0.02020) = 34.3$ days
- 4. $0.02 + 0.08 \times 9 + 0.90 \times 1.5 = 2.090$; $\sigma_{t+1} = 1.446\%$
- A fall of 3% raises the variance by 39%; half of the excess is gone after 34.3 trading days, about 7 weeks.

A2: A More Reactive Market [Proposed]

Solution discussed in class

Task

- $\omega = 0.05$, $\alpha = 0.12$, $\beta = 0.85$, $\mu = 0$; today $\sigma_t^2 = 2.0$ and $r_t = +1.5\%$. Model: A1.
- 1. Compute the persistence, the long-run variance and the annualised long-run volatility.
- 2. Compute the half-life.
- 3. Compute σ_{t+1}^2.
- 4. Write EWMA with $\lambda = 0.94$ as a GARCH(1,1) and say which of the quantities in 1–2 exist for it.
- Report: four numbers and two sentences.

Back Exercise list

A3: Forecasts and VaR 1% [Solved]

Task

- The model of A1, with $\sigma^2_{t+1} = 2.090$ and Normal innovations.
- 1. Compute $E_t[\sigma^2_{t+5}]$ and $E_t[\sigma^2_{t+10}]$.
- 2. Compute the 10-day variance and volatility.
- 3. Compare with the square-root-of-time rule $\sqrt{10}\,\sigma_{t+1}$ and with $\sqrt{10\,\bar{\sigma}^2}$.
- 4. Compute the one-day and the 10-day VaR 1%.
- Report: six numbers and one sentence.

Solution

- 1. $1 + 0.98^4(2.090 - 1) = 1 + 0.9224 \times 1.09 = 2.005$; $1 + 0.98^9 \times 1.09 = 1.909$
- 2. $10 \times 1 + 1.09 \times (1 - 0.98^{10})/0.02 = 10 + 1.09 \times 9.146 = 19.97$; volatility 4.47%
- 3. $\sqrt{10 \times 2.090} = 4.57\%$ (too high), $\sqrt{10\,\bar{\sigma}^2} = \sqrt{10 \times 1} = 3.16\%$ (too low)
- 4. $2.326 \times 1.446 = 3.36\%$; $2.326 \times 4.47 = 10.40\%$
- The high variance decays slowly: the 10-day risk lies between the two simple rules, closer to the first.

Back Exercise list

A4: Forecasts with Student-t Innovations [Proposed]

Solution discussed in class

Task

- The model of A2, with $\sigma^2_{t+1} = 2.020$ and standardised Student-t innovations with $\nu = 5$ ($t_5^{-1}(0.01) = -3.365$). Model: A3.
- 1. Compute $E_t[\sigma^2_{t+10}]$ and the 10-day variance.
- 2. Compute the quantile $q_{0.01}$ of the standardised t.
- 3. Compute the one-day and the 10-day VaR 1% with t and with Normal innovations.
- Report: six numbers and one sentence.

A5: The ARCH(1) Likelihood, Step by Step [Solved]

Task

- ARCH(1) with $\omega = 0.5$, $\alpha = 0.5$, $\mu = 0$, Normal z_t; observed: $r_1, \ldots, r_4 = 0.5;\ -1.2;\ 2.0;\ -0.3$.
- 1. Compute σ_2^2, σ_3^2 and σ_4^2.
- 2. Compute ℓ_2, ℓ_3, ℓ_4 and the conditional log-likelihood.
- 3. Compute the unconditional variance and the kurtosis of this ARCH(1).
- Report: seven numbers and one sentence.

Solution

- 1. $0.5 + 0.5 \times 0.25 = 0.625$; $0.5 + 0.5 \times 1.44 = 1.220$; $0.5 + 0.5 \times 4 = 2.500$
- 2. $\ell_2 = -\frac{1}{2}[1.8379 + \ln 0.625 + 1.44/0.625] = -1.836$; $\ell_3 = -2.658$; $\ell_4 = -1.395$; sum -5.889
- 3. $0.5/(1 - 0.5) = 1.0$; $K = 3 \times 0.75/0.25 = 9.0$
- Day 3 ($r_3 = 2$ after a calm day) contributes least: a large return when σ_t^2 is small is "unlikely".

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A6: Good and Bad News [Proposed]

Solution discussed in class

Task

- GJR-GARCH: $\omega = 0.02$, $\alpha = 0.03$, $\gamma = 0.12$, $\beta = 0.90$; EGARCH: $\omega = 0$, $\alpha = 0.12$, $\gamma = -0.08$, $\beta = 0.98$; in both $\sigma_t^2 = 1$, $\mu = 0$. Model: A1.
- 1. Compute the GJR σ_{t+1}^2 after $\varepsilon_t = -2$ and after $\varepsilon_t = +2$, and their ratio.
- 2. Compute the GJR persistence and half-life.
- 3. Compute the EGARCH σ_{t+1}^2 for the same two shocks (use $E|z| = 0.7979$).
- Report: six numbers and one sentence.

B1: GARCH(1,1)-t for the DAX [Solved]

- **Question**: how persistent is DAX volatility, and how far is the DAX from its long-run volatility today?
- **Inputs**: DAX closes since 2000, daily log returns in %
- **Tasks**
  1. Estimate GARCH(1,1) with Student-t innovations and report the parameters with robust standard errors.
  2. Compute the persistence, the half-life and the annualised long-run volatility, and compare it with the sample volatility.
  3. Report the peaks of the annualised conditional volatility in 2008 and in 2020, and its value on the last day.
  4. Draw the returns since 2018 with $\pm 2\hat{\sigma}_t$ bands.
  5. Interpretation: should a risk manager use the long-run volatility or the current one for tomorrow?
- **Report**: a table of five parameters, five numbers, the chart and two sentences
- Notebook: section B1

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B1: Solution [Solved]



	μ	ω	α	β	ν
estimate	0.081	0.0206	0.097	0.895	7.08
robust SE	0.012	0.0049	0.011	0.012	0.62

- $T = 6\,786$; $\alpha + \beta = 0.992$, half-life $\ln 0.5/(-0.00784) = 88.5$ days; long-run volatility 25.9% against 22.4% in the sample
- Peaks: 80% (2008), 82% (2020); last day: 13.0%; 5.2% of the days fall outside $\pm 2\hat{\sigma}_t$
- Interpretation: the current one: VaR for tomorrow needs $\sigma_{t+\mathbf{1}}$; the long-run level matters only for long horizons, and it is imprecise because $1 - \alpha - \beta = 0.0078$ is small

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B2: BET, Banca Transilvania and Bitcoin [Proposed]

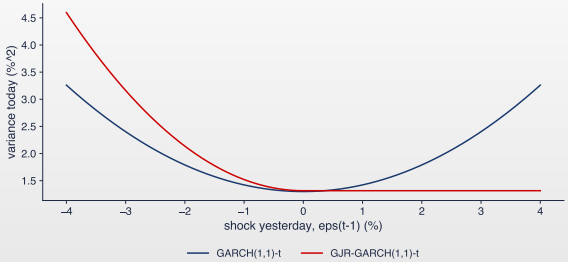
- **Question**: which of the BET, Banca Transilvania and Bitcoin has the most persistent volatility? Model: B1.
- **Inputs**: BET since 2000, TLV since 2010 (without 30–31 May 2016), Bitcoin since 2014; daily log returns in %
- **Tasks**
  1. Estimate GARCH(1,1)-t for each series and report α, β and ν.
  2. Compute the persistence, the half-life and the long-run volatility where they exist.
  3. Compare the long-run with the sample volatility (annualise Bitcoin with 365 days).
  4. Interpretation: what does an estimate $\hat{\alpha} + \hat{\beta} = 1$ tell a risk manager?
- **Report**: one table and two sentences
- Notebook: section B2

B3: The Leverage Effect in the S&P 500 [Solved]

- **Question**: do falls raise the volatility of the S&P 500 more than rises of the same size?
- **Inputs**: S&P 500 closes since 2000, daily log returns in %
- **Tasks**
  1. Estimate GARCH(1,1)-t and GJR-GARCH(1,1)-t; report γ with its robust t statistic.
  2. Compute the LR statistic of GJR against GARCH, its p-value and the two BIC values.
  3. Run the sign-bias test on the standardised residuals of GARCH-t.
  4. Draw both news impact curves and compute the GJR variance after shocks of -2% and $+2\%$.
  5. Interpretation: how does the leverage effect change the VaR on the day after a fall?
- **Report**: four statistics with p-values, two variances, the chart and two sentences
- Notebook: section B3

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B3: Solution [Solved]



- GJR: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.03$), $\hat{\beta} = 0.881$; LR $= 235.7$ (p < 0.001); BIC 18169.0 $\rightarrow$ 17942.2
- Sign bias on GARCH-t residuals: $\chi^2(3) = 40.7$ (p < 0.001), sign $t = 3.56$
- GJR variance after -2%: 2.136; after $+2\%$: 1.315; ratio 1.62
- Interpretation: only falls raise the variance ($\hat{\alpha} = 0$); after a fall the VaR is about $\sqrt{1.62}$ times the VaR after an equal rise; a symmetric GARCH understates risk after bad days

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B4: Asymmetry on the BVB and in Bitcoin [Proposed]

- **Question**: is there a leverage effect in the BET, in Banca Transilvania and in Bitcoin? Model: B3.
- **Inputs**: BET since 2000, TLV since 2010, Bitcoin since 2014; daily log returns in %
- **Tasks**
  1. Estimate GJR-GARCH(1,1)-t and EGARCH(1,1)-t; report both γ with robust t statistics.
  2. Compute the LR test of GJR against GARCH and the change in BIC.
  3. Compute the GJR ratio of the variances after -2% and $+2\%$.
  4. Interpretation: would you use an asymmetric model for each of the three? Why?
- **Report**: one table and two sentences
- Notebook: section B4

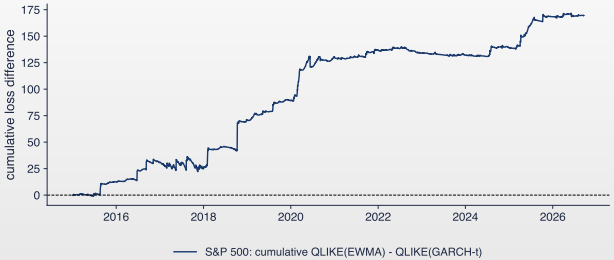
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B5: GARCH against EWMA for the S&P 500 [Solved]

- **Question**: does GARCH(1,1)-t forecast tomorrow's variance of the S&P 500 better than EWMA, and is its VaR 1% exceeded on 1% of days?
- **Inputs**: S&P 500 since 2000; forecasts for every day since 2 January 2015
- **Tasks**
  1. Re-estimate GARCH(1,1)-t every 250 days on all data up to that day and compute the one-day forecasts h_t; compute EWMA with $\lambda = 0.94$.
  2. Compute the mean QLIKE of both models and the Diebold–Mariano t statistic.
  3. Count the exceedances of the GARCH-t VaR 1% and of the EWMA-Normal VaR 1% and compare with the expected number.
  4. Draw the cumulative difference of the losses.
  5. Interpretation: which model would you give to a risk manager, and what would you still fix?
- **Report**: four numbers, two counts, the chart and two sentences
- Notebook: section B5

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B5: Solution [Solved]



- 2 944 days; mean QLIKE: GARCH-t 0.761, EWMA 0.819; DM: mean difference -0.058, $t = -3.69$ ($p < 0.001$): GARCH-t is better
- VaR 1% exceedances: GARCH-t 49 (1.66%), EWMA-Normal 65 (2.21%); expected about 29
- Interpretation: GARCH-t, but both VaR models are exceeded too often; add the leverage effect (B3) and test the exceedances formally (Chapter 10)

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B6: GARCH against EWMA for the DAX and the BET [Proposed]

- **Question**: does the result of B5 hold for the DAX and for the BET? Model: B5.
- **Inputs**: DAX and BET since 2000; forecasts for every day since January 2015
- **Tasks**
  1. Compute the out-of-sample GARCH(1,1)-t and EWMA forecasts as in B5.
  2. Compute the mean QLIKE of both models and the DM test.
  3. Count the VaR 1% exceedances of both models.
  4. Interpretation: for which index does GARCH help most, and why?
- **Report**: one table and two sentences
- Notebook: section B6

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C1: Is Bitcoin Volatility Becoming Equity-Like? [Proposed]

- **Question**: has Bitcoin volatility moved closer to the volatility of the S&P 500 since 2021?
- **Inputs**: Bitcoin and S&P 500 daily log returns, 2015–2019 and 2021–2026; models: B1, B3
- **Tasks**
  1. Estimate GJR-GARCH(1,1)-t for both series in both periods and report the persistence, γ and ν.
  2. Compute the correlation of the logarithms of the two GARCH-t volatilities on common days in each period.
  3. List the events of 2020–2024 that could explain a change.
  4. Interpretation: how would you separate a lasting change from a change caused by one crisis?
- **Report**: a table and a plan for a project
- Notebook: section C1

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C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to interpret a GARCH(1,1)-t fit of the S&P 500 (daily, since 2000): $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\nu} = 6.26$. The answer:
- `(a) alpha + beta = 0.995, so a volatility shock disappears within about a week.`
- `(b) The long-run variance is omega/(1 - alpha) = 0.018, an annual volatility of 2.1%.`
- `(c) A GJR fit gives gamma = 0.205 > 0: good news raises volatility more than bad news.`
- `(d) Ljung-Box Q(10) of the squared standardised residuals = 13.6, p = 0.19: the GARCH-t model is correct.`
- `(e) With Student-t innovations the one-day VaR 1% is 2.326 times sigma(t+1).`
- `(f) EWMA with lambda = 0.94 is a GARCH(1,1) with omega = 0, alpha = 0.06, beta = 0.94.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- $\alpha + \beta$ decides everything long-run: half-life, long-run variance, the shape of the forecasts
- Multi-day risk is the sum of the daily variance forecasts, not $\sqrt{H}$ times today's volatility
- Heavy-tailed innovations change the VaR quantile; asymmetry changes the VaR after bad days
- Compare forecasts out of sample (QLIKE, DM), not only in sample (AIC, BIC)
- An AI answer is a draft: check the formula of the long-run variance, the sign of γ and the quantile

After the Seminar

- Lecture 9 develops each topic of today: ARCH and GARCH, estimation, innovations, asymmetry, diagnostics, forecasts, VaR
- C1 can grow into a team project: rolling windows, Ethereum and gold as controls
- Reading: Franke, Härdle & Hafner (2019), Ch. 13; exercises in Borak, Härdle & López-Cabrera (2013), Ch. 13; Tsay (2010), Ch. 3

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